Biometrics ??, 1–23 DOI: 10.1111/j.1541-0420.2005.00454.x
March 2026

Multiomics Tissue Segmentation via Spatially-Informed Nested Biclustering Methods

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Summary: Matrix-Assisted Laser Desorption/Ionization (MALDI) Mass Spectrometry Imaging (MSI) is a powerful technique for spatially resolved molecular profiling and cancer biomarker discovery. Recent advances, including a novel multiomics workflow, enable multiple rounds of MSI on the same tissue section, extracting diverse molecular classes, such as lipids, N-glycans, and peptides, while preserving spatial resolution. This innovation is particularly valuable for studies with limited tissue sample availability, such as rare diseases or small tumors. However, the resulting data are high-dimensional and spatially structured, with the various molecular types sharing the same pixel grid. To address these challenges, we propose Poseidon, a Bayesian nonparametric nested biclustering model that simultaneously segments the common tissue pixels and clusters molecular signals within each molecular class, leveraging the shared spatial structure. A separately exchangeable framework is first considered, and then extended to handle spatial data via Markov random fields. For scalability, we implement an efficient mean-field variational inference algorithm tailored for multi-dataset analysis. After validating the efficacy of our method on simulated scenarios, we demonstrate its applicability in a real-world case study, where multiomics measurements were performed on kidney tissue affected by clear cell renal cell carcinoma. The nested, hierarchical structure of Poseidon, combined with its principled inferential framework, allows the extraction of meaningful biological insights, such as clear tissue segmentation and biomarker detection.

Key words: Clear cell renal cell carcinoma; Dirichlet process; Potts model; Shared atoms nested models; Variational inference.

This paper has been submitted for consideration for publication in *Biometrics*

1. Introduction

Over the last decade, *Matrix-Assisted Laser Desorption/Ionization* (MALDI) Mass Spectrometry Imaging (MSI) has emerged as a key technology for *in situ* molecular profiling and cancer biomarker discovery. The technique measures *mass spectra*, i.e., the molecular abundances of a given molecular class (e.g., lipids) across locations of a biological sample (Rohner et al., 2005). In MALDI-MSI, a tissue section is first coated with a chemical compound called *matrix*, which isolates a specific class of molecules and prepares them for detection. The slide is then divided into a grid of virtual pixels, typically 10–50 μm in size, each irradiated by a laser beam. The instrument separates the detected molecules - the *analytes* - according to their mass-to-charge ratio (m/z), producing a mass spectrum that records the abundance (intensity) of each m/z signal.

Recent advances in MSI have improved both detection precision and experimental reproducibility, expanding spatial proteomics and enabling finer characterization of tumor microenvironments (Krestensen et al., 2023; Zhang et al., 2024). A particularly important development was introduced by Denti et al. (2022), who proposed a *multiomics workflow* enabling multiple MSI acquisitions on the *same* tissue section. By sequentially applying and removing different *matrices*, this approach extracts analyte intensities from multiple molecular classes (e.g., lipids, N-glycans, and peptides) while minimizing tissue degradation. As a result, each molecular layer yields a distinct MSI dataset measured over a *common spatial grid*. This strategy is particularly valuable when clinical samples are limited, such as in rare diseases or small tumors, as it maximizes molecular information obtained from a single tissue section.

However, these experimental advances introduce new analytical challenges. Multiomics MSI data are inherently high-dimensional and spatially structured, and the novel *multiomics workflow* further increases complexity by generating nested data structures. The pipeline outputs multiple MSI matrices (one for each molecular class), which share the same spatial grid of pixels (columns), but differ in the analytes they measure (rows). A core objective in this context is tissue segmentation, which involves identifying biologically meaningful subregions of a sample based on their spectral characteristics, to uncover subtle molecular variations that may be missed by morphological inspection alone.

Current approaches typically rely on linear and nonlinear dimensionality reduction techniques to extract simplified representations of MSI data (see Zhvansky et al., 2021). Building on these representations, clustering methods, most notably k-means and bisecting k-means, remain standard in the field, while probabilistic frameworks such as Gaussian mixture models have also shown promising results (Guo et al., 2019; Prasad et al., 2022). See also Tuck et al. (2022) for a comprehensive review. Recently, more sophisticated models have been introduced to address the intricate characteristics of omics data, incorporating spatial structure and domain-specific features. These include Bayesian spatial clustering approaches (e.g., Zhao et al., 2021; Sottosanti and Risso, 2023) or graph-based community detection algorithms (e.g., Kim et al., 2022; Hu et al., 2024). However, most of these methods are designed for single-matrix inputs and therefore fail to capture the shared spatial structure, nested organization, and molecular heterogeneity that arise across different analyte types.

In this paper we propose Poseidon, a novel Bayesian nonparametric (BNP) method for the analysis of multiomics MSI data based on *nested biclustering*, which simultaneously segments spatial pixels and clusters molecular signals within each tissue subregion. Our approach integrates the multiple MSI datasets while explicitly accounting for the shared spatial grid and underlying dependencies across pixels. In particular, Poseidon builds on the concept of separate exchangeability (see Rebaudo et al., 2026, and the references therein), providing a principled foundation for modeling complex nested structures in high-dimensional omics data. This framework naturally induces a two-layer partitioning structure, similar to that discussed in Lee et al. (2013). Our proposed method contributes in three key ways. First, by jointly clustering pixels and m/z values, the model identifies signal groups with elevated or suppressed activation, effectively distinguishing biologically meaningful peaks from background noise and enabling the extraction of latent molecular feature representations that support the spatial segmentation of the tissue. Second, it accommodates the nested structure of multiomics MSI data by integrating multiple molecular classes into a unified framework, allowing comparisons across analyte types measured on the same tissue. Finally, to ensure scalability and efficient inference in high-dimensional settings, we implement a tailored variational approximation to the posterior distribution, drawing on recent advances in mean-field variational inference for hierarchical, nested Bayesian models (Blei et al., 2017; D'Angelo and Denti, 2026).

1.1 *Clear Cell Renal Cell Carcinoma MALDI-MSI Data*

Our application focuses on MALDI-MSI measurements obtained from a human tissue sample diagnosed with clear cell renal cell carcinoma (ccRCC), shown in panel A of Figure 1 (Denti et al., 2026). The tissue is annotated as follows: the green region indicates the *healthy cortex*, the blue outlines the *capsule*, and the red highlights the *tumoral nodule*. A more detailed segmentation is provided in Figure S.2 of the Supplementary Materials. Renal cell carcinoma is among the ten most frequently diagnosed cancers worldwide, with the clear-cell subtype accounting for roughly 75% of cases and for most related mortality (Hsieh et al., 2017). Prognosis is often poor due to high recurrence rates and resistance to conventional chemotherapy and radiotherapy. Investigating the phenotypic characteristics of cells within these tumor environments may reveal novel molecular pathways and potential therapeutic strategies (Hsieh et al., 2018).

Using the multiomics pipeline of Denti et al. (2022), mass spectra were acquired for three molecular classes on the same renal tissue section: *lipids, N-glycans*, and *tryptic peptides*.

These classes were selected for their complementary biological roles. Lipids are major components of cell membranes, provide energy for cell growth, and act as signaling molecules that may contribute to drug resistance (Bian et al., 2021; Fu et al., 2021; Shafiee et al., 2020). N-glycans are among the most common protein modifications and play a key role in directing proteins to specific cellular compartments (Stanley et al., 2022). Proteins themselves regulate diverse processes, including cell growth, tissue architecture, and cellular replication (Gobena et al., 2024). In this protocol, proteins are enzymatically cleaved by trypsin to generate tryptic peptides, enabling more detailed proteomic analysis. Together, these three molecular layers provide a complementary view of tumor biology and can enhance biomarker discovery when analyzed within an integrated spatial framework, as demonstrated in previous spatial multiomic studies (Balluff et al., 2019).

In the preparatory phase, the raw spectra extracted with MALDI underwent routine pre-processing steps to account for physiological, instrumental, and analytical variability. Specifically, the spectra followed the biological five-step protocol described in Capitoli et al. (2025), summarized in Section S.1 of the Supplementary Materials. These pre-processed signals were then stored in three separate matrices, corresponding to lipids, N-glycans, and peptides. In each matrix, columns represent spatial pixels (shared across matrices). At the same time, rows correspond to analyte-specific m/z signals (which differ by molecular class). Each matrix contains approximately 100 rows and over 4,000 columns. Figure 1 provides a visual overview of the data. Panels B, C, and D show the normalized median abundance of each molecular class across the tissue section. These images highlight molecular heterogeneity that is not always apparent from histology alone. Panels F, G, and H illustrate spectral profiles for a single pixel (highlighted in panel E), showing the abundance levels across m/z indices for each molecular class. Note that normalization was performed within each molecular class to enhance visualization, so intensities are not directly comparable across the three panels.

By applying our proposed method to these data, we aim to identify molecular signatures and spatial subregions within the renal tissue that distinguish tumor from non-tumor areas, offering insights into the heterogeneity of ccRCC. Moreover, the biclustering approach aligns with our broader goal of reducing data complexity and delivering interpretable, low-dimensional summaries to support pathologists in downstream analyses.

The rest of the paper is organized as follows. In Section 2, we introduce our biclustering model, first outlining the core method for a single molecular layer and then extending it to jointly handle multiple analytes. Section 3 presents the variational inference algorithm used for posterior estimation. Section 4 summarizes the simulation studies reported in Section S.4 of the Supplementary Materials. In Section 5, we apply the model to the ccRCC dataset and analyze the resulting spatial and molecular patterns. Section 6 concludes.

[Figure 1 about here.]

2. A finite-infinite separately exchangeable model for MALDI-MSI data

In line with the data structure outlined in Section 1.1, we consider a collection of T datasets $\boldsymbol{\mathcal{Y}} = \{\boldsymbol{Y}^{(t)}\}_{t=1}^{T}$, one for each molecular level. Then, $\boldsymbol{Y}^{(t)}$ denotes the MALDI-MSI abundance matrix of the t-th molecular class, which is characterized by $N^{(t)}$ rows, representing the specific analytes, and J columns, representing the pixels. We remark that a crucial aspect of the data is that the number of columns is the same across all the datasets. Our goal is to estimate a nested random partition to perform image segmentation, i.e., inducing a *shared partition* of the J pixels driven by the distributional characteristics of the molecular abundance within each location across datasets and, at the same time, inducing an idiosyncratic, *dataset-specific partition* of the signals, different in each dataset. Figure 2 summarizes the objective of our model. In what follows, we begin by discussing our model for the single-dataset case ($T = 1$) and then extend it to the multiple-dataset scenario.

[Figure 2 about here.]

2.1 *The single-dataset case: Pose*

Let us focus on a single dataset, say $\boldsymbol{Y}$, with N rows and J columns. From a modeling perspective, we see our data as organized into distinct but related groups defined by the pixels, or columns. This setting is typical of nested designs, an area of research that has thrived in recent years within the Bayesian literature (see, for example, Rodríguez et al., 2008; Camerlenghi et al., 2019; Denti et al., 2023; Balocchi et al., 2021). Nested models aim to simultaneously cluster groups and observations; for matrix data, this corresponds to clustering both the columns and the entries. Most of these nested models were proposed under the hypothesis of *partial exchangeability*, where data are assumed to be exchangeable within the different subpopulations. Consequently, the joint probability distribution of observations remains unchanged under any permutation of data points within the same subpopulation. Under this assumption, the probability of clustering matrix entries does not depend on their row identity, and these models completely disregard the information conveyed by the specificity of each row, as Rebaudo et al. (2026) have recently underlined. Indeed, this feature is problematic for us, as each row of our matrix has the same meaning (i.e., it represents the same m/z value) across all the columns. Using a partially exchangeable model in our framework would mean clustering the pixels that have similar abundance distributions by simply considering the proportion of m/z that have high vs low activation levels, regardless of *which* m/z achieve these activation levels. This might lead to an undesirable situation where, for example, two pixels are clustered together because they both have 5% of the signals that are highly activated, even though the high activation is achieved by different sets of m/z in the two pixels, leading to an important loss of information.

To overcome this issue, similarly to Rebaudo et al. (2026), we start by considering a nested model in a *separately exchangeable* setting, so that the identity of each row is preserved across the groups. We will introduce a set of column-specific clusters and, for each of them, a set of row-specific clusters, which are named, respectively, column clusters (CCs) and row clusters (RCs). Specifically, for each column $j = 1, \ldots, J$, let C_j denote the CC membership label, with $C_j \in \{1, \ldots, K\}$. Given the vector of CC allocations $\boldsymbol{C} = (C_1, \ldots, C_J)$, separate exchangeability implies that in each CC all the entries in a row are restricted to belong to the same RC. In other words, for each row $i = 1, \ldots, N$ and *every possible CC* k, we consider a RC membership label $R_{i,k} \in \{1, \ldots, L\}$ that is common to all entries (i, j) whose columns are in the same CC characterized by $C_j = k$. Such RC allocation is considered for all possible CC realizations, with $\boldsymbol{R}_k = (R_{1,k}, \ldots, R_{N,k})$ for all k. The RC assignment variables $R_{i,k}$ are categorical random variables with CC-specific parameters, such that $\mathbb{P}\left[R_{i,k} = l | \boldsymbol{\omega}_k\right] = \omega_{l,k}$, where $\boldsymbol{\omega}_k = (\omega_{1,k}, \ldots, \omega_{L,k})$ is a vector of probabilities, for all k. In the basic model specification, which we will extend later, we also consider the CC assignment to be independent categorical random variables with probabilities $\boldsymbol{\pi}$, i.e., $\mathbb{P}\left[C_j = k | \boldsymbol{\pi}\right] = \pi_k$. Let $f(y \mid \theta)$ denote a suitable parametric density for the data, and let $\{\theta_l^*\}_{l=1}^L$ be a collection of RC-specific parameters shared across all CCs. The conditional density of entry $y_{i,j}$, given the column-cluster assignment $C_j = k$ and the row-cluster assignment $R_{i,k} = l$, is assumed to be $f(y_{i,j} \mid \theta_{i,j})$, with $\theta_{i,j} = \theta^*_{R_{i,C_j}}$. Hence, whenever $C_j = k$ and $R_{i,k} = l$, the corresponding parameter satisfies $\theta_{i,j} = \theta_l^*$. Finally, we assume that $\theta_l^* \overset{i.i.d.}{\sim} H$, for $l = 1, \ldots, L$, where H is a diffuse base measure. Thus, given all the vectors of categorical probabilities $\boldsymbol{\pi}$ and $\{\boldsymbol{\omega}_k\}_{k=1}^K$,

we can express our model in terms of the cluster assignment variables $R_{i,k}$ and C_j as

$$\theta_l^* \overset{i.i.d.}{\sim} H,$$
$$C_j|\boldsymbol{\pi} \sim \mathrm{Cat}_K(\boldsymbol{\pi}), \qquad R_{i,k}|\boldsymbol{\omega}_k \sim \mathrm{Cat}_L(\boldsymbol{\omega}_k), \tag{1}$$
$$y_{i,j}|C_j, R_{i,C_j}, \{\theta_l^*\}_l \overset{ind.}{\sim} f(y_{i,j}|\theta^*_{R_{i,C_j}}).$$

Note that the matrix entries $y_{i,j}$ associated to the same RC l, i.e., the observations belonging to the set $\{y_{i,j} : R_{i,C_j} = l\}$, are assumed to be exchangeable; in other words, they are modeled as conditionally i.i.d. from $f(\cdot|\theta_l^*)$.

Remark 1: Drawing a parallel with the partially exchangeable nested model, we could describe this model by introducing column-specific random measures G_j, assumed to be distributed according to a discrete random measure $Q = \sum_{k\geqslant 1} \pi_k \delta_{G_k^*}$. The discreteness of Q induces an implicit clustering on the columns, and we could define $C_j = k$ when $G_j = G_k^*$. For every possible CC k, we could consider row-specific parameters $\theta_{i,k}$, drawn i.i.d. from G_k^*. When the G_k^*'s are also discrete, with $G_k^* = \sum_{l\geqslant 1} \omega_{l,k}\delta_{\theta_l^*}$, a partition is also induced on the $\theta_{i,k}$ parameters, leading to the definition of the RC assignments $R_{i,k}$.

Essentially, we cluster together different m/z if they have similar activation levels (a trait captured by the kernel parameters $\{\theta_l^*\}_{l=1}^L$), and we group together different pixels if they have similar activation profiles, as represented by the vector of cluster assignments of the m/z. Note that according to the specification in (1), the distributional law of the data matrix $\boldsymbol{Y}$ is invariant to permutations of entire rows and columns. Moreover, given the nested nature of the model, the clustering of signals is affected by the CCs, as the cluster assignment of a given signal remains the same for all pixels belonging to the same CC.

Which prior distributions are selected for $\{\boldsymbol{\omega}_k\}_{k=1}^K$ (the *observational weights*) and $\boldsymbol{\pi}$ (the *distributional weights*) represents a crucial modeling choice. In Rebaudo et al. (2026), the authors consider a fully nonparametric approach, allowing both L and K to be infinite, and adopt stick-breaking (SB) Dirichlet process (DP) representations for both weights, inducing a *common atom* structure. Here, we consider a finite-infinite version of the model, resulting in a *shared atom* model, following D'Angelo and Denti (2026), and we restrict L to be finite, while K is allowed to be infinite. Specifically, we adopt sparse, symmetric Dirichlet distributions with parameters $\boldsymbol{b}_0 = b_0 \mathbb{1}_L$ for the row mixture weights, in order to empty useless mixture components. Thus, for all k, $\boldsymbol{\omega}_k \sim \text{Dirichlet}_L(\boldsymbol{b}_0)$. In the following, we will show how this simple change has a serious impact on the prior probability of coclustering rows. Similarly to Rebaudo et al. (2026), we could assume that $\boldsymbol{\pi}$ is drawn from a SB process with concentration parameter α, denoted with GEM(α), which is equivalent to setting $\pi_1 = v_1$ and $\pi_k = v_k \prod_{g<k}(1 - v_g)$ where $v_j \overset{i.i.d.}{\sim} \text{Beta}(1, \alpha)$ for all $j \geqslant 1$.

However, the assumption of prior independence in the CC membership labels $\boldsymbol{C}$, which would be implied under separate exchangeability, could be naive in the context of spatial data. In fact, the columns in our dataset represent pixels of an image, which are likely to be spatially correlated. Therefore, we need to elicit a distribution that allows clustering of the columns, while conveying their spatial information. Specifically, we assume that the clustering assignment for the generic pixel j is influenced by the cluster memberships of its adjacent pixels. This structure can be recovered using discrete-state Markov Random Fields (MRF, Besag, 1986), also known as Potts models when the number of states considered is larger than two. Here, we adopt the BNP-MRF prior introduced by Lü et al. (2020), which extends the classic BNP models for clustering using a MRF. In detail, this distribution assumes that $p(C_j \mid \boldsymbol{C}_{-j}, \boldsymbol{\pi}, \beta) \propto \pi_{C_j} \cdot \exp(\beta \sum_{q \in \partial_j} \mathbb{1}_{\{C_q = C_j\}})$, where $\boldsymbol{\pi} \sim \text{GEM}(\alpha)$, and ∂_j denotes the set of neighbors of unit j, i.e., the set of pixels adjacent to pixel j. In this work, we employ a second-order neighborhood system (queen adjacency) for each pixel, as it yields a more flexible and realistic characterization of local spatial interactions compared to first-order (rook) adjacency (Besag, 1986). The quantity β, known as the inverse temperature, governs the level of influence that the neighboring cluster memberships have on a pixel, and can be treated as a fixed hyperparameter or modeled as a random quantity. This conditional distribution can be equivalently stated as a joint distribution for the whole vector of cluster memberships $\boldsymbol{C}$, as

$$p(\boldsymbol{C} \mid \boldsymbol{\pi}, \beta) = \frac{1}{\mathcal{K}_{\boldsymbol{C}}(\boldsymbol{\pi}, \beta)} \left(\prod_{j=1}^{J} \pi_{C_j} \right) \cdot \exp \left[\beta \sum_{j=1}^{J} \sum_{q \in \partial_j} \mathbb{1}_{\{C_q = C_j\}} \right], \tag{2}$$

where $\mathcal{K}_{\boldsymbol{C}}(\boldsymbol{\pi}, \beta)$ is an intractable normalizing constant. Therefore, we redefine the prior distribution for the CC membership $\boldsymbol{C}$ as (2), and denote it with $\boldsymbol{C}|\boldsymbol{\pi}, \beta \sim \text{BNP-MRF}(\boldsymbol{\pi}, \beta)$.

We refer to the model in (1) with the CC prior redefined as in (2) as the *Potts Over Separate Exchangeability* (Pose) model. Note that under Pose, the model is no longer separately exchangeable, due to the MRF term of the prior for $\boldsymbol{C}$, and the simple discrete measure representation using Q is no longer possible.

2.1.1 *Prior coclustering properties under Pose.* A key distinction arises between common-atom and shared-atom models. As discussed in D'Angelo and Denti (2026), combining an infinite set of atoms with a common ordering across random measures and stick-breaking priors induces possibly undesired dependencies among the observational weights in partially exchangeable common-atom models. This typically results in high prior coclustering probabilities, which may adversely affect posterior inference. This issue persists even under separate exchangeability (see also Supplementary Section S.4.1). For the Pose model, it is therefore instructive to examine how the prior coclustering probability between two entries, $\mathbb{P}\left[\theta_{i,j} = \theta_{i',j'}\right]$, changes when $\boldsymbol{\omega}_k \sim \text{Dirichlet}_L(\boldsymbol{b}_0)$ (shared atoms), rather than the common atoms induced by a DP. Proposition 1 of Lü et al. (2020) provides the predictive distribution of the BNP-MRF process for general Gibbs-type priors. Specialized to the DP and the case of two observations, we can derive the probability of ties between two groups, equal to $\mathbb{P}\left[G_j = G_{j'}\right] = e^{\beta} / \left(\alpha + e^{\beta}\right)$. We can then prove the following proposition. The proof is deferred to Section S.2.1 of the Supplementary Materials.

Proposition 2.1: The prior coclustering probability between two entries according to the shared atoms Pose model with $\boldsymbol{\omega}_k \sim \text{Dirichlet}_L(\boldsymbol{b}_0)$ for all k and $\boldsymbol{\pi} \sim GEM(\alpha)$ is

$$\mathbb{P}\left[\theta_{i,j} = \theta_{i',j'}\right] = \frac{1}{\alpha + e^{\beta}} \left[e^{\beta} \left(\frac{1 + b_0}{1 + Lb_0} \right)^{1 - \mathbb{1}_{i=i'}} + \frac{\alpha}{L} \right].$$

In the same setting, if instead we assume $\boldsymbol{\omega}_k \sim GEM(\nu)$ for all k, we have

$$\mathbb{P}\left[\theta_{i,j} = \theta_{i',j'}\right] = \frac{1}{\alpha + e^{\beta}} \left[e^{\beta} \left(\frac{1}{1 + \nu} \right)^{1 - \mathbb{1}_{i=i'}} + \frac{\alpha}{2\nu + 1} \right].$$

A graphical depiction of these probabilities is shown in Supplementary Figures S.3-S.5. Consistent with previous findings (D'Angelo and Denti, 2026), the prior coclustering probability is higher under the common atoms model, highlighting its stronger a priori grouping and risk of spurious coclustering. Distinguishing between same-row ($i = i'$) and different-row ($i \neq i'$) entries is important, as when two groups (columns j and j') are clustered, their rows automatically share the same partition. The difference between models is most apparent for different rows, where clustering behavior depends on the observational weight hyperparameter (b_0 for shared atoms, ν for common atoms). For same-row pairs, coclustering under shared atoms is independent of b_0, but still depends on ν for common atoms. As the spatial regularization parameter β increases, model differences diminish and clustering probabilities converge.

2.2 *The multiple-dataset case: Poseidon*

Given the definition of Pose for a single molecule dataset, we can simply extend the biclustering framework to multiple datasets. Consider again the collection of T datasets $\boldsymbol{\mathcal{Y}} = \{\boldsymbol{Y}^{(t)}\}_{t=1}^{T}$, each characterized by $N^{(t)}$ rows and J columns. We now consider T collections of row partitions $\boldsymbol{\mathcal{R}}^{(t)} = \{\boldsymbol{R}_k^{(t)}\}_{k=1}^{K}$ (one for each dataset $t = 1, \ldots, T$), with a dataset-specific row partition $\boldsymbol{R}_k^{(t)}$ for each CC k. We also denote with $\boldsymbol{\theta}^{*(t)}$ the collection of cluster-specific parameters for each dataset $t = 1, \ldots, T$, and let $\boldsymbol{\theta}^* = (\boldsymbol{\theta}^{*(1)}, \ldots, \boldsymbol{\theta}^{*(T)})$. Finally, we assume the existence of one partition of the columns $\boldsymbol{C} = (C_1, \ldots, C_J)$, shared by all the datasets, resulting in a common image segmentation. Given these partitions, we assume the measurements are conditionally independent across datasets, setting $p(\boldsymbol{\mathcal{Y}} \mid \{\boldsymbol{\mathcal{R}}^{(t)}\}_{t=1}^{T}, \boldsymbol{C}, \boldsymbol{\theta}^*) = \prod_{t=1}^{T} p(\boldsymbol{Y}^{(t)} \mid \boldsymbol{\mathcal{R}}^{(t)}, \boldsymbol{C}, \boldsymbol{\theta}^{*(t)})$. Then, we can express the model in the following form:

$$\begin{gathered}\theta_l^{*(t)} \overset{i.i.d.}{\sim} H^{(t)}, \quad t = 1, \ldots, T \\ \boldsymbol{C}|\boldsymbol{\pi}, \beta \sim \text{BNP-MRF}(\boldsymbol{\pi}, \beta), \qquad R_{i,k}^{(t)}|\boldsymbol{\omega}_k^{(t)} \sim \text{Cat}_L(\boldsymbol{\omega}_k^{(t)}), \quad t = 1, \ldots, T \\ y_{i,j}^{(t)}|C_j, \{R_{i,k}^{(t)}\}_k, \{\theta_l^{*(t)}\}_l \overset{ind.}{\sim} f(y_{i,j}^{(t)}|\theta^{*(t)}_{R^{(t)}_{i,C_j}}), \quad t = 1, \ldots, T,\end{gathered} \tag{3}$$

with $\boldsymbol{\pi} \sim GEM(\alpha)$, $\boldsymbol{\omega}_k^{(t)} \sim \text{Dirichlet}_L(\boldsymbol{b}_0)$, $\forall t, k$, and each $H^{(t)}$ is a diffuse base measure. One can also specify a prior for the concentration parameter of the distributional DP, e.g., $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$, and a prior for the inverse temperature, e.g., $\beta \sim \mathcal{U}(0, B)$.

We call this model *Potts Over Separate Exchangeability for the Integration of Different Omics Nested data* (Poseidon). Thanks to the hierarchical structure used in defining Poseidon, the model has the flexibility to model the molecule-specific datasets $\boldsymbol{Y}^{(t)}$, using dataset-specific parameters $\boldsymbol{\theta}^{*(t)}$; nevertheless, it allows for sharing information across these datasets to infer the shared partition of the columns, leveraging the signal provided by the different molecules (see also Supplementary Section S.4.4). For our application, we will specialize model (3) to fit a mixture of Gaussian distributions. As the first step, we need to transform the positive measurements of the original abundance matrices, here denoted with $\boldsymbol{Z}^{(t)}$, into real values as

$$y_{i,j}^{(t)} = \Phi^{-1}\left(\frac{z_{i,j}^{(t)} - \underline{z}^{(t)}}{\bar{Z}^{(t)} - \underline{z}^{(t)}}\right), \quad \text{for} \quad i = 1, \ldots, N; \quad j = 1, \ldots, J, \quad t = 1, \ldots, T, \tag{4}$$

where $\Phi^{-1}(\cdot)$ represents the inverse c.d.f. of a Normal distribution, $\underline{z}^{(t)} = \min_{i,j}(z_{i,j}^{(t)}) - \epsilon$, and $\bar{Z}^{(t)} = \max_{i,j}(z_{i,j}^{(t)}) + \epsilon$, where a small $\epsilon > 0$ ensures the arguments remain within $(0, 1)$ to prevent divergence. To complete our model specification, exploiting conjugacy, we finally specify Normal-Inverse Gamma priors for the base measures $H^{(t)}$, $t = 1, \ldots, T$. Thus, for all l, we assume $\theta_l^{*(t)} = (\mu_l^{*(t)}, \sigma_l^{2*(t)})$ where $\mu_l^{*(t)} \mid \sigma_l^{2*(t)} \sim \text{N}(m_0^{(t)}, \sigma_l^{2*(t)}/k_0^{(t)})$ and $\sigma_l^{2*(t)} \sim \text{IG}(c_0^{(t)}, d_0^{(t)})$. Recommended values for the hyperparameters are $(m_0^{(t)}, k_0^{(t)}, c_0^{(t)}, d_0^{(t)}) = (0, 0.01, 3, 2)$ for all t. We also set $\boldsymbol{b}_0 = 10^{-4} \cdot \mathbf{1}_L$ to ensure sparse, overfitted finite mixtures across the rows (Malsiner-Walli et al., 2016), effectively estimating the number of non-empty clusters. Finally, we set $a_\alpha = b_\alpha = 1$ and $B = 2$. A detailed discussion of the selected hyperparameters is provided in Supplementary Section S.3.3.

3. Posterior inference via variational Bayes

We perform posterior inference using a tailored variational Bayes (VB) algorithm to ensure an efficient estimation of Poseidon on our high-dimensional MALDI-MSI data. In short, let $\boldsymbol{\Theta} = \left(\{\boldsymbol{\mathcal{R}}^{(t)}\}_t, \boldsymbol{C}, \boldsymbol{\theta}^*, \boldsymbol{\pi}, \{\boldsymbol{\omega}_k^{(t)}\}_{t,k}, \alpha, \beta\right)$ represent the set of all the parameters in our model. After a family of approximating densities $\mathcal{Q}$ is specified over $\boldsymbol{\Theta}$, we seek to find the member $q^*(\boldsymbol{\Theta})$ of $\mathcal{Q}$ that minimizes the Kullback-Leibler divergence ($\mathcal{D}_{KL}$) with the true posterior distribution. Note that minimizing this divergence is equivalent to maximizing the Evidence Lower Bound (ELBO), defined as $ELBO(q) = \mathbb{E}\left[\log p\left(\boldsymbol{\Theta}, \boldsymbol{\mathcal{Y}}\right)\right] - \mathbb{E}\left[\log q\left(\boldsymbol{\Theta}\right)\right]$. The variational family $\mathcal{Q}$ is typically characterized by variational parameters $\boldsymbol{\lambda}$. The optimization problem is thus reduced to finding the optimal parameters $\boldsymbol{\lambda}^* = \arg\min_{\boldsymbol{\lambda}} \mathcal{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\lambda}}(\boldsymbol{\Theta}) \parallel p(\boldsymbol{\Theta} \mid \boldsymbol{\mathcal{Y}})\right)$. We opt for a mean-field variational approximation, which chooses $\mathcal{Q}$ assuming the parameters are mutually independent. To find $\boldsymbol{\lambda}^*$, we resort to a Coordinate Ascent Variational Inference (CAVI) procedure, detailed in Section S.3 of the Supplementary Materials. The single-dataset case (Pose model) is easily recovered when $T = 1$. Almost all the CAVI steps can be performed in closed form, as the corresponding full conditionals are known distributions. The only steps that require approximations are the updates for the parameters of the distributions of v_k's and the inverse temperature β. For the former approximation, we mimic Lü et al. (2020) and assume $\beta = 0$. For the latter, we proceed as delineated in Supplementary Section S.3.1. To fit the model, we initialize our CAVI algorithm using different starting points, stopping the computations when the relative increment in the ELBO is below a certain threshold, such as 10^{-8}. Once all the runs are performed, we retain the estimate that obtained the highest value of the ELBO. The details for the computation of the ELBO components are in Section S.3.2 of the Supplementary Materials.

4. Simulations

We conducted several simulation studies to evaluate the performance of our proposal. All the studies are deferred to the Supplementary Materials. First, Section S.4.1 demonstrates why shared atoms are preferred over common atoms in this context. In Section S.4.2, we validate Pose's ability to recover biclusters accurately, and in Section S.4.3 we show that Pose achieves better cluster recovery than state-of-the-art competitors across datasets with varying degrees of spatial dependence. In Section S.4.4, we show that nested clustering with Poseidon is driven by the shared signal rather than corrupted by noise. Finally, Section S.4.5 analyzes computational scalability, presenting the average per-iteration runtime (in seconds) as J, T, and N vary.

5. ccRCC image segmentation and molecular signal extraction

We fit Poseidon to the ccRCC dataset described in Section 1.1 after applying the transformation in (4). To estimate the model, we set the upper bound for the number of CCs and RCs to $K = 30$ and $L = 40$, respectively. We run the algorithm using 250 different starting points on a Linux machine with 32 GB of memory. The average running time is 3.2 minutes, with a standard deviation of 1.2 minutes. We summarize the main results below; additional details are provided in Section S.5 of the Supplementary Materials.

5.1 *Tissue segmentation*

To evaluate our proposed multiomic approach, we first examine the spatial partitions generated by Poseidon, obtained by integrating information across the three molecular classes (Figure 3, panel E). Our proposal reduces modality-specific noise and yields a unified, coherent, and biologically meaningful view of the tissue's molecular landscape. By imposing a shared column partition, Poseidon produces an interpretable solution with a concise number of clusters (15 CCs). To provide an indirect measure of goodness-of-fit of the estimation, we compute the root mean squared error (RMSE) by aggregating the discrepancies between the observed data values and the corresponding estimated means within each bicluster. For Poseidon, the resulting average RMSE is 0.029. The estimated densities for each CC are provided in the Supplementary Materials. Importantly, Poseidon reveals multiomic alterations in both the tumor microenvironment and adjacent histologically normal tissue, features that may be difficult to capture through standard histopathological evaluation due to limited morphological differences (Figure 1, panel A).

Figure 3 also reports the results of different Pose models estimated on the individual datasets: lipids (panel A), N-glycans (panel B), and peptides (panel C), and allows us to highlight the underlying biological architectures across which Poseidon enables integration. Indeed, single-molecule segmentation reveals that N-glycans and peptides are closely linked, as N-glycans bind to specific peptide sequences and act as labels that direct proteins to their cellular destinations. In the healthy region of the tissue (right side of the section), the clustering of N-glycans and peptides shows greater concordance than the lipid modality, identifying a non-concentric large-scale organization of the healthy cortex. Within the tumor nodule, however, their clustering patterns diverge, likely reflecting disrupted glycosylation processes in cancer (Wallace et al., 2024). Both of these layers successfully identify the mixed-grade tumor (G2–G3) region. In contrast, individual lipid-based segmentation reveals concentric layers within the healthy region, a pattern also detected by Poseidon (panel E). This suggests the presence of metabolic or inflammatory gradients that operate largely independently of the protein–glycosylation architecture. Lipids also highlight distinct regions within the tumor, including areas associated with inflammatory infiltration and hemorrhagic features (panel F), as well as concentric layers surrounding the tumor nodule that extend into the healthy region, which are less evident in the N-glycan and peptide segmentations. Notably, in the upper portion of the tumor nodule, Poseidon consistently delineates a localized region (Tumor G2–G3 in panel F), which is more prominently captured in the N-glycan and peptide modalities than in the lipid layer.

Finally, we compare our approach against a model estimated on a stacked dataset combining all three molecular classes. While stacking offers a straightforward alternative for data integration, it is generally considered a less principled approach (see Supplementary Section S.4.4 for further discussion). This yields a slightly better fit (RMSE: 0.022) but results in a larger number of CCs (19 CCs), which hinders interpretability. The stacked analysis treats all signals as originating from a single dataset. As a result, some tissue areas appear more fragmented, especially where peptide and N-glycan layers suggest distinct spatial organizations. Furthermore, the resulting clusters predominantly reflect the lipid-driven concentric structure in the healthy region. This suggests that the lipid modality drives the stacked segmentation, while additional molecular-specific patterns may be less distinctly represented. Overall, while similar spatial features are observed with stacking, an integrative method such as Poseidon produces a cleaner segmentation and more effectively captures the distinct characteristics of each molecular modality described above.

[Figure 3 about here.]

5.2 *Results on* m/z *in specific CCs*

To illustrate the type of inference enabled by the Poseidon formulation, we focus on three CCs associated with biologically meaningful tissue regions (left panels of Figure 4). Two correspond to distinct areas within the tumor nodule (CC-10 and CC-6, top and bottom panels), while the central panel (CC-12) represents a healthy region of the kidney cortex. CC-10 corresponds to a mixed-grade tumor region, annotated by the pathologist as intermediate between *grade 2* and *grade 3*, whereas CC-6 contains hemorrhagic tissue with tumor cells classified as *grade 2*. Distinguishing between these tumor areas is challenging even for experienced pathologists, given their subtle morphological differences. Poseidon's multiomic integration, in contrast, enables a spatially resolved characterization of molecular behavior across these regions.

The right panels of Figure 4 display the posterior median of each row assigned to the CCs. Dots, colored according to their m/z partition, highlight how specific molecular signals contribute to each molecular class, allowing the identification of the signals that most strongly characterize each region. N-glycans show broadly similar patterns across the tumor clusters (CC-10 and CC-6), consistent with well-documented cancer-associated glycan alterations, while the benign region (CC-12) displays a distinct profile. This difference is particularly evident in the mid-m/z range (approximately 1688–1926), where the tumor clusters exhibit a characteristic low–high–low triplet pattern across adjacent signals, whereas the benign one shows a steadily increasing trend. The similarity between tumor clusters reflects the tendency of cancer tissues to share glycosylation alterations, even when their histological appearances differ. In contrast, lipidomic signals reveal bigger differences between tumor regions. Marked variation is observed in the higher m/z range, likely reflecting distinct metabolic states in healthy and tumor tissue. CC-6 shows particularly elevated abundance of small lipids in the 606–619 m/z range, consistent with the hemorrhagic nature of this region. Interestingly, these lower m/z lipid signals exhibit more similar behavior between CC-10 and the healthy CC-12, possibly reflecting inflammatory processes present in both tumor-adjacent tissue and portions of the normal cortex.

Peptides, instead, tend to be more abundant in the healthy region. In the mid-range of the spectrum (approximately m/z 700–1500), several peptide signals form a group of peaks that are consistently elevated in CC-12 while remaining weaker in the tumor clusters. Among these are peptides corresponding to nuclear proteins such as m/z 944.53 (*histone H2A*) and m/z 1116.56 (putatively identified as *Heterochromatin Protein 1-binding protein 3*), recognized tumor markers in the MSI literature (Denti et al., 2024). While proteomic alterations in cancer have been extensively studied, the functional roles of N-glycans and lipids remain less explored, particularly with respect to their spatial organization across tissue regions (Vasseur and Guillaumond, 2022; Wallace et al., 2024). Our findings provide a foundation for future investigations into the spatially resolved roles of these molecular layers in disease processes. Complementary comments on the other CCs are reported in Supplementary Section S.5.2.

[Figure 4 about here.]

6. Discussion

We introduced Poseidon, a BNP method for the joint analysis of multi-molecule MALDI-MSI data, designed to effectively integrate the heterogeneous information provided by different spatially resolved multiomics experiments. By combining a BNP-MRF prior with a separately exchangeable biclustering framework, we could simultaneously segment a tissue and cluster molecular features across omics classes while preserving spatial dependencies. This principled integration improved over both single-dataset and naively combined approaches, as demonstrated in our analysis of lipids, N-glycans, peptides, and stacked datasets. The improvements observed, both in terms of biologically meaningful tissue segmentation and the discovery of plausible molecular biomarkers, underscore the importance of integrated models for spatially structured molecular data.

This work opens up several promising research directions. First, one can consider different likelihood specifications, allowing for a more flexible modeling of the data without resorting to pre-processing transformations. Second, alternative spatially informed priors could be explored as an alternative to the MRF, which is known to pose computational challenges. For example, thresholded Gaussian processes or spatially dependent SB priors (e.g., Linderman et al., 2015) may offer more flexible and scalable alternatives. Finally, incorporating stochastic variational inference could enable scalability to larger datasets, facilitating information sharing across multiple tissues from the same patient collected over time, or across tissues from different patients with the same disease. This would potentially allow for integrated, one-step differential analyses across space, time, and individuals.

Acknowledgment

FD acknowledges support by the STARS@UNIPD 2025 program, with a contribution from the Fondazione Cassa di Risparmio di Padova e Rovigo.

Supplementary Materials

Web Appendices and Figures referenced in Sections 1-5 are available with this paper at the Biometrics website on Oxford Academic. Furthermore, the code and data retrieval instructions required to reproduce the results are hosted in the GitHub repository at https://github.com/Fradenti/poseidon-maldi.

Data Availability

The data (Denti et al., 2026) mentioned in Sections 1 and 5 of this paper are publicly available on the *Bicocca Open Archive Research Data* and can be accessed at `https://doi.org/10.17632/gxz9kjj2r9.1`.

References

Balluff, B., Buck, A., Martin-Lorenzo, M., Dewez, F., Langer, R., McDonnell, L. A., Walch, A., and Heeren, R. M. (2019). Integrative clustering in mass spectrometry imaging for enhanced patient stratification. *PROTEOMICS–Clinical Applications* **13,** 1800137.

Balocchi, C., George, E. I., and Jensen, S. T. (2021). Clustering Areal Units at Multiple Levels of Resolution to Model Crime in Philadelphia. *ArXiv preprint* .

Besag, J. (1986). On the Statistical Analysis of Dirty Pictures. *Journal of the Royal Statistical Society. Series B (Methodological)* **48,** 259–302.

Bian, X., Liu, R., Meng, Y., Xing, D., Xu, D., and Lu, Z. (2021). Lipid metabolism and cancer. *Journal of Experimental Medicine* **218,** 1–34.

Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. (2017). Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association* **112,** 859–877.

Camerlenghi, F., Dunson, D. B., Lijoi, A., Prünster, I., and Rodríguez, A. (2019). Latent Nested Nonparametric Priors (with Discussion). *Bayesian Analysis* **14,** 1–54.

Capitoli, G., van Abeelen, K. C. J., Piga, I., L'Imperio, V., Nobile, M. S., Besozzi, D., and Galimberti, S. (2025). Well Begun Is Half Done: The Impact of Pre-Processing in MALDI Mass Spectrometry Imaging Analysis Applied to a Case Study of Thyroid Nodules. *Stats* **8,** 1–14.

D'Angelo, L. and Denti, F. (2026). A Finite-Infinite Shared Atoms Nested Model for the Bayesian Analysis of Large Grouped Data Sets. *Bayesian Analysis* **21,** 105 – 138.

Denti, F., Balocchi, C., Capitoli, G., and Denti, V. (2026). Dataset: Poseidon Data ccRCC Application. *Bicocca Open Archive Research Data* .

Denti, F., Camerlenghi, F., Guindani, M., and Mira, A. (2023). A Common Atoms Model for the Bayesian Nonparametric Analysis of Nested Data. *Journal of the American Statistical Association* **118,** 405–416.

Denti, V., Capitoli, G., Piga, I., Clerici, F., Pagani, L., Criscuolo, L., Bindi, G., Principi, L., Chinello, C., Paglia, G., et al. (2022). Spatial multiomics of lipids, n-glycans, and tryptic peptides on a single ffpe tissue section. *Journal of Proteome Research* **21,** 2798–2809.

Denti, V., Greco, A., Alviano, A. M., Capitoli, G., Monza, N., Smith, A., Pilla, D., Maggioni, A., Ivanova, M., Venetis, K., et al. (2024). Spatially resolved molecular characterization of noninvasive follicular thyroid neoplasms with papillary-like nuclear features (niftps) identifies a distinct proteomic signature associated with ras-mutant lesions. *International Journal of Molecular Sciences* **25,** 13115.

Fu, Y., Zou, T., Shen, X., Nelson, P. J., Li, J., Wu, C., Yang, J., Zheng, Y., Bruns, C., Zhao, Y., Qin, L., and Dong, Q. (2021). Lipid metabolism in cancer progression and therapeutic strategies. *MedComm* **2,** 27–59.

Gobena, S., Admassu, B., Kinde, M. Z., and Gessese, A. T. (2024). Proteomics and its current application in biomedical area: Concise review. *The Scientific World Journal* **2024,** 4454744.

Guo, D., Bemis, K., Rawlins, C., Agar, J., and Vitek, O. (2019). Unsupervised segmentation of mass spectrometric ion images characterizes morphology of tissues. *Bioinformatics* **35,** i208–i217.

Hsieh, J. J., Le, V., Cao, D., Cheng, E. H., and Creighton, C. J. (2018). Genomic classifications of renal cell carcinoma: A critical step towards the future application of personalized kidney cancer care with pan-omics precision. *The Journal of pathology* **244,** 525–537.

Hsieh, J. J., Purdue, M. P., Signoretti, S., Swanton, C., Albiges, L., Schmidinger, M., Heng, D. Y., Larkin, J., and Ficarra, V. (2017). Renal cell carcinoma. *Nature reviews Disease primers* **3,** 1–19.

Hu, Y., Rong, J., Xu, Y., Xie, R., Peng, J., Gao, L., and Tan, K. (2024). Unsupervised and supervised discovery of tissue cellular neighborhoods from cell phenotypes. *Nature Methods* **21,** 267–278.

Kim, J., Rustam, S., Mosquera, J. M., Randell, S. H., Shaykhiev, R., Rendeiro, A. F., and Elemento, O. (2022). Unsupervised discovery of tissue architecture in multiplexed imaging.

Nature Methods **19,** 1653–1661.

Krestensen, K. K., Heeren, R. M., and Balluff, B. (2023). State-of-the-art mass spectrometry imaging applications in biomedical research. *Analyst* **148,** 6161–6187.

Lee, J., Müller, P., Zhu, Y., and Ji, Y. (2013). A nonparametric Bayesian model for local clustering with application to proteomics. *Journal of the American Statistical Association* **108,** 775–788.

Linderman, S. W., Johnson, M. J., and Adams, R. P. (2015). Dependent multinomial models made easy: Stick-breaking with the pólya-gamma augmentation. In *Advances in Neural Information Processing Systems 28*, pages 3456–3464.

Lü, H., Arbel, J., and Forbes, F. (2020). Bayesian nonparametric priors for hidden Markov random fields. *Statistics and Computing* **30,** 1015–1035.

Malsiner-Walli, G., Frühwirth-Schnatter, S., and Grün, B. (2016). Model-based Clustering Based on Sparse Finite Gaussian Mixtures. *Statistics and Computing* **26,** 303–324.

Prasad, M., Postma, G., Franceschi, P., Buydens, L. M., and Jansen, J. J. (2022). Evaluation and comparison of unsupervised methods for the extraction of spatial patterns from mass spectrometry imaging data (msi). *Scientific Reports* **12,** 15687.

Rebaudo, G., Lin, Q., and Mueller, P. (2026). Separate Exchangeability as Modeling Principle in Bayesian Nonparametrics. *Statistical Science* .

Rodríguez, A., Dunson, D. B., and Gelfand, A. E. (2008). The nested dirichlet process. *Journal of the American Statistical Association* **103,** 1131–1154.

Rohner, T., Staab, D., and Stoeckli, M. (2005). Maldi mass spectrometric imaging of biological tissue sections. *Mechanisms of Ageing and Development* **126,** 177–185.

Shafiee, M. N., Ortori, C. A., Barrett, D. A., Mongan, N. P., Abu, J., and Atiomo, W. (2020). Lipidomic biomarkers in polycystic ovary syndrome and endometrial cancer. *International Journal of Molecular Sciences* **21,** 1–15.

Sottosanti, A. and Risso, D. (2023). Co-clustering of spatially resolved transcriptomic data. *The Annals of Applied Statistics* **17,** 1444–1468.

Stanley, P., Moremen, K. W., Lewis, N. E., Taniguchi, N., and Aebi, M. (2022). *N-Glycans*. Cold Spring Harbor Laboratory Press, Cold Spring Harbor (NY), 4th edition.

Tuck, M., Grélard, F., Blanc, L., and Desbenoit, N. (2022). Maldi-msi towards multimodal imaging: Challenges and perspectives. *Frontiers in Chemistry* **10,** 1–11.

Vasseur, S. and Guillaumond, F. (2022). Lipids in cancer: a global view of the contribution of lipid pathways to metastatic formation and treatment resistance. *Oncogenesis* **11,** 46.

Wallace, E. N., West, C. A., McDowell, C. T., Lu, X., Bruner, E., Mehta, A. S., Aoki-Kinoshita, K. F., Angel, P. M., and Drake, R. R. (2024). An n-glycome tissue atlas of 15 human normal and cancer tissue types determined by maldi-imaging mass spectrometry. *Scientific Reports* **14,** 489.

Zhang, H., Lu, K. H., Ebbini, M., Huang, P., Lu, H., and Li, L. (2024). Mass spectrometry imaging for spatially resolved multi-omics molecular mapping. *npj Imaging* **2,** 20.

Zhao, E., Stone, M. R., Ren, X., Guenthoer, J., Smythe, K. S., Pulliam, T., Williams, S. R., Uytingco, C. R., Taylor, S. E. B., Nghiem, P., Bielas, J. H., and Gottardo, R. (2021). Spatial transcriptomics at subspot resolution with bayesspace. *Nature Biotechnology* **39,** 1375–1384.

Zhvansky, E., Sorokin, A., Shurkhay, V., Zavorotnyuk, D., Bormotov, D., Pekov, S., Potapov, A., Nikolaev, E., and Popov, I. (2021). Comparison of dimensionality reduction methods in mass spectra of astrocytoma and glioblastoma tissues. *Mass Spectrometry* **10,** A0094.

Received October 2007. *Revised February* 2008. *Accepted March* 2008.

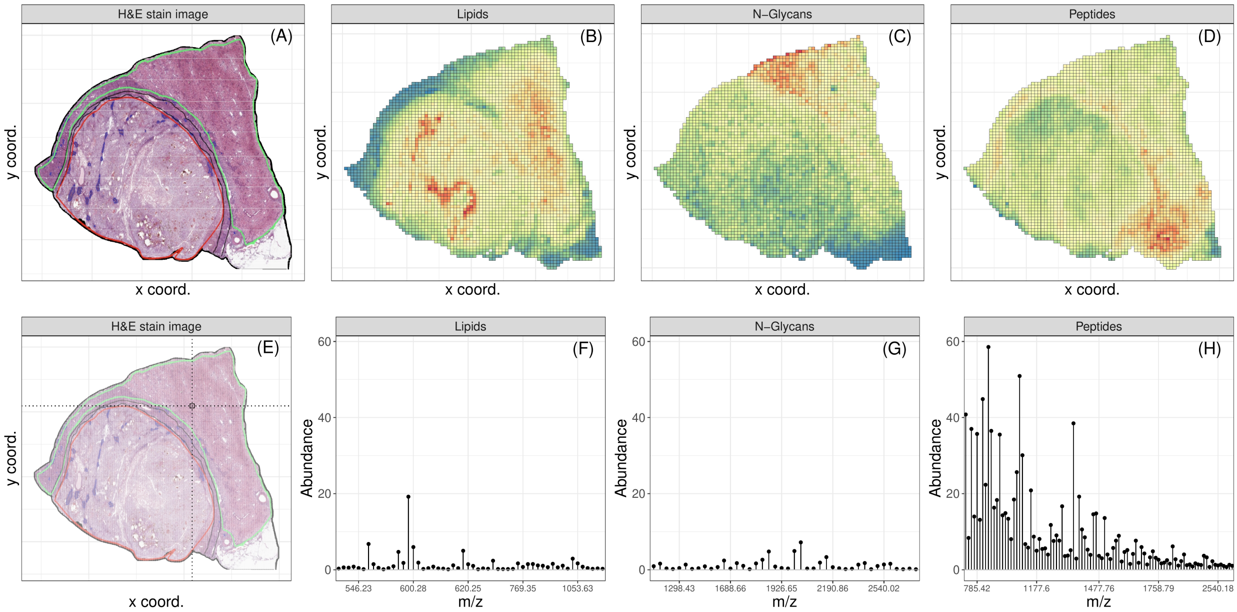

Figure 1. The top row displays the Hematoxylin-and-Eosin stained image of the ccRCC tissue used in the analysis (A) and heatmaps displaying the (normalized) median abundance across the tissue sample for lipids, N-glycans, and peptides (B-D). The bottom row reports, for a specific highlighted pixel (E), abundances of the signals for the different molecular classes (F-H).

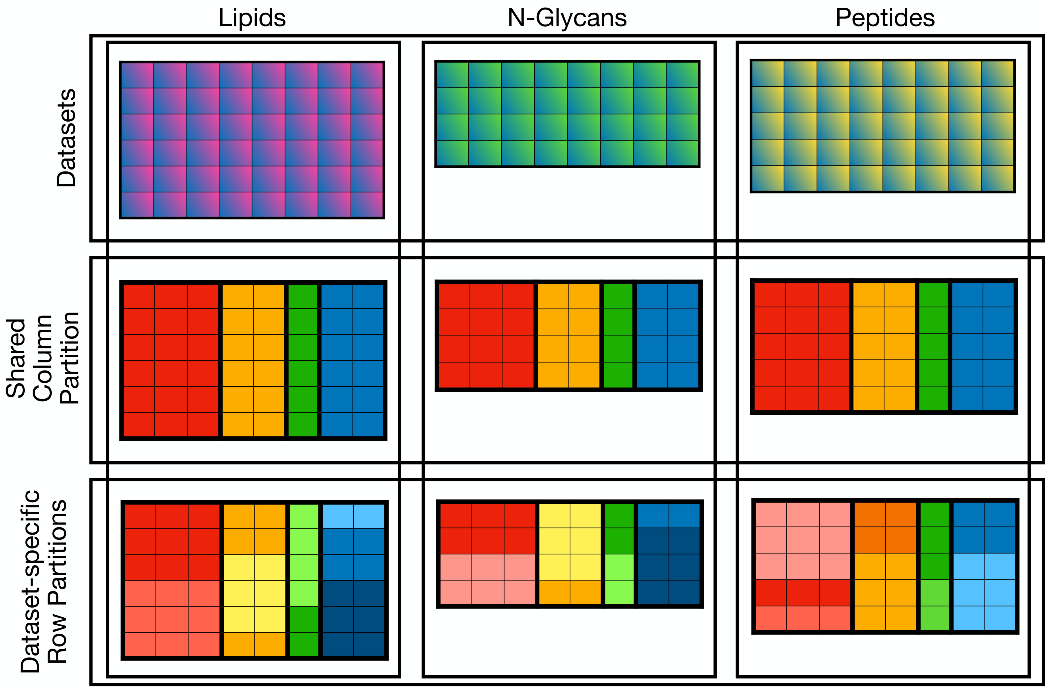

Figure 2. A diagram summarizing the partition structure we aim to recover. The partition over the columns is shared across datasets. The partition over the rows is dataset-specific.

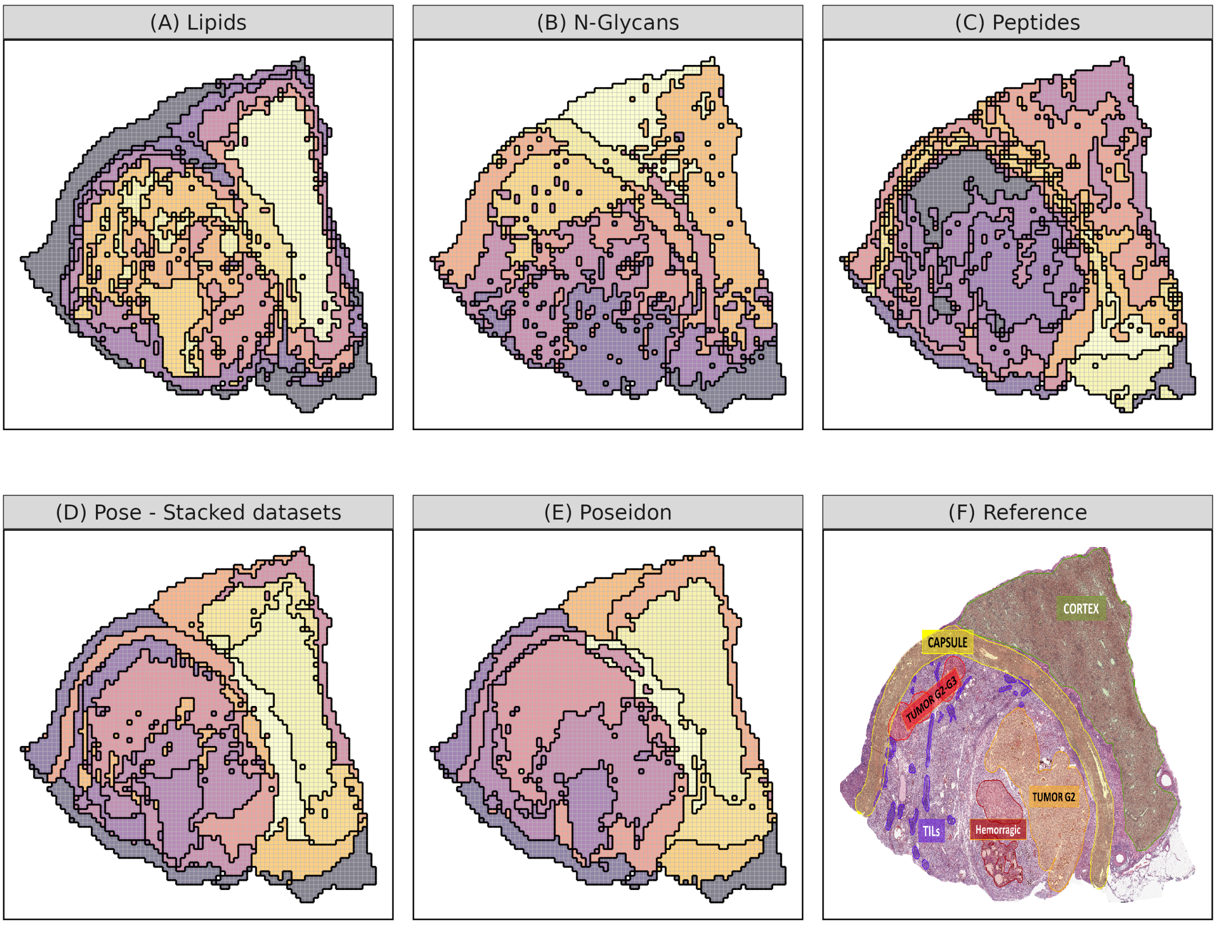

Figure 3. Tissue segmentation obtained by fitting single-dataset Pose models to the separate molecules (panels A–C) and the stacked datasets (panel D). These results are compared with Poseidon's multi-dataset approach (panel E) and the original ccRCC tissue annotated by a pathologist (panel F).

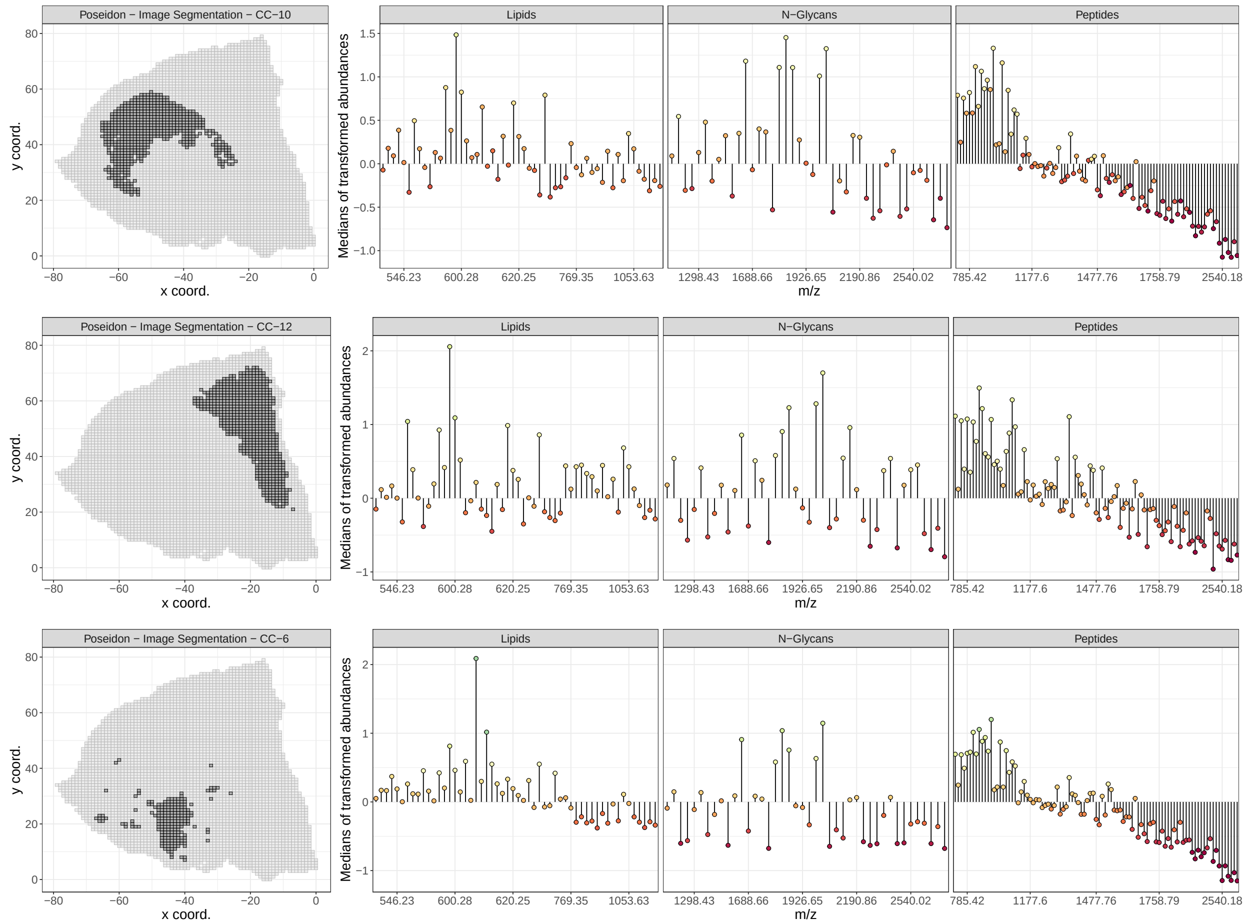

Figure 4. Spatial and signal characterization of three representative regions of interest identified within the tissue by Poseidon. The left panels show the pixels corresponding to each estimated CC, highlighting the localization patterns across the sample. Right panels display the corresponding posterior median of the analytes' rescaled abundances in each CC, segmented by molecular class. Therefore, each panel illustrates the relative median intensities of m/z features contributing to the cluster-specific molecular profile, and these are colored according to the RC assignments.